COST ANALYSIS REPORT

STM32-Based DC Motor Speed & Direction Control System (H-Bridge MOSFET Driver)

Prepared for Electronics Engineering Workshop Project Submission | Currency: Indian Rupees (Rs.) | Basis: Proteus Schematic Bill of Materials

1. Introduction

This report presents an estimated cost analysis for the real-world implementation of the DC Motor Speed and Direction Control System designed and simulated in Proteus Design Suite. The design uses an STM32F103C6 ("Blue Pill") microcontroller to generate PWM signals that drive a discrete MOSFET H-bridge (IRF9530N high-side P-channel MOSFETs and IRLZ44N low-side N-channel MOSFETs, gate-driven through 2N2222 transistors) to control the speed and rotational direction of a 12V brushed DC motor. Motor speed is set through a potentiometer and direction is selected using two push-button switches. The cost estimates in this report are based on commercially available electronic components and current approximate Indian retail/wholesale market prices as of 2026. Where an exact component match was not commercially identifiable, the closest commonly available equivalent part has been used and is clearly marked as an approximate estimate.

2. Component Cost Table

Sl. No.	Component Name	Qty	Est. Unit Price (Rs.)	Total Cost (Rs.)
1	STM32F103C6 "Blue Pill" MCU Board	1	280.00	280.00
2	IRF9530N P-Channel Power MOSFET	2	35.00	70.00
3	IRLZ44N N-Channel Logic-Level MOSFET	2	30.00	60.00
4	2N2222 NPN Bipolar Transistor	2	3.00	6.00
5	Resistor - 10 kOhm, 1/4W	4	1.00	4.00
6	Resistor - 100 kOhm, 1/4W	2	1.00	2.00
7	Resistor - 1 kOhm, 1/4W	1	1.00	1.00
8	Resistor - 100 Ohm, 1/4W	3	1.00	3.00
9	Potentiometer - 1 kOhm Linear	1	15.00	15.00
10	Push Button Switch (SPST, Tactile)	2	3.00	6.00
11	DC Motor (12V, Brushed) *	1	120.00	120.00
12	Battery - 12V DC Supply Pack *	1	350.00	350.00
	Total Electronic Components Cost			**Rs. 917.00**

** Exact commercial part not uniquely identifiable from the schematic; price is estimated using the closest commonly available equivalent component.*

3. Category-wise Cost Distribution

Category	Included Components	Subtotal (Rs.)
Semiconductors	STM32F103C6 board, IRF9530N x2, IRLZ44N x2, 2N2222 x2	416.00
Passive Components	Resistors (10k x4, 100k x2, 1k x1, 100R x3), Potentiometer 1k	25.00
Electromechanical Components	DC Motor 12V, Push Button Switches x2	126.00
Power Supply Components	12V Battery Pack	350.00
Connectors	None identified as discrete schematic parts (see Section 4 for wiring/terminal costs)	0.00
Miscellaneous	Not applicable - no additional schematic components	0.00
	Total	**Rs. 917.00**

4. Additional Manufacturing Costs

Beyond the electronic components themselves, converting the schematic into a working physical prototype requires PCB fabrication and assembly-related consumables. Estimated costs below are for a single small prototype board sourced from typical Indian PCB fabrication and electronics retail suppliers.

Item	Estimated Cost (Rs.)
PCB Fabrication (2-layer, prototype quantity)	350.00
Soldering Materials (solder wire, flux, wick)	100.00
Connecting Wires (single-core / hook-up wire)	80.00
Screw Terminals / Berg Connectors / Battery Clip	120.00
Heat Sinks for TO-220 MOSFETs (with thermal paste)	150.00
Miscellaneous Assembly Cost (enclosure, fasteners, contingency)	250.00
Total Additional Manufacturing Cost	**Rs. 1,050.00**

5. Grand Total

Total Electronic Components Cost	Rs. 917.00
Additional Manufacturing Cost	Rs. 1,050.00
Estimated Final Project Cost	**Rs. 1,967.00**

6. Cost Optimization Suggestions

1. Purchase resistors, capacitors and small hardware in bulk kits/packs rather than as individual pieces to reduce the effective per-unit price.
2. Source MOSFETs, transistors and the microcontroller board from local wholesale electronics markets (e.g., Lamington Road, Mumbai or SP Road, Bangalore) instead of retail online stores for lower prices.

3. Use a single-sided, hand-soldered prototype PCB instead of a fabricated double-sided board for a one-off student project to significantly cut PCB fabrication cost.
4. Omit dedicated heat sinks if the motor operating current stays well below the MOSFET current rating and duty cycle is low; rely on PCB copper pour for heat dissipation instead.
5. Reuse push buttons, connectors, wires and mounting hardware salvaged from earlier lab kits/projects instead of purchasing new ones.
6. Use an existing lab bench power supply or a repurposed 12V adapter for demonstration purposes instead of buying a dedicated battery pack.
7. Order all components together from a single local supplier or online store in one consolidated order to save on shipping and handling charges.
8. Consider a generic/clone STM32 "Blue Pill" board (widely available at lower cost) instead of an original branded board, since functionality for this project does not require the original.

Disclaimer

All prices quoted in this report are approximate estimates in Indian Rupees (Rs.), based on typical retail and wholesale market rates for commercially available components as of 2026. Actual prices may vary depending on the supplier, purchase quantity, brand, geographic location, and prevailing market conditions. This report is intended for academic and project-planning purposes only and should not be treated as a formal quotation.